Lecture 7: Bayesian Games, Adverse Selection and Auctions

Detailed solutions to the 24 items of `Listas/pratica-lecture-07.md`
Economics of Organizations and Incentives (MPE Insper)

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Each item follows the list's order and numbering. Statements are in `pratica-lecture-07.md` and are not repeated. Every number and equilibrium claim is asserted in `check_E1.py` (items 1–8), `check_E2.py` (9–18) and `check_E3.py` (19–24), run from this folder. Notation: a type $t_i \in T_i$ is drawn by Nature from a common prior; a strategy is a function $s_i : T_i \to A_i$; a profile s^* is a Bayesian Nash equilibrium (BNE) if every type t_i of every player maximizes his expected payoff given his posterior over t_{-i} and the others' strategies (Gibbons §3.1.B, pp. 149–150; Tadelis §12.1, pp. 246–252). The unit is estimated: there are no Lecture 7 slides yet, so all references are to the books.

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Lecture 7 (estimated) — Bayesian games, adverse selection and auctions

Item 1 (item 1 of the combined list)**: Gibbons Problem 3.1** Foundations · difficulty 1/5

Source: Gibbons Problem 3.1, page 169. The statement is reproduced verbatim in the list (item 1); it is not repeated here.

(a) What is asked. Define a static Bayesian game, a pure strategy in it, and a pure-strategy Bayesian Nash equilibrium.

(b) Method. Gibbons §3.1.B (pp. 149–151); Tadelis §12.1 (pp. 246–252). Information is incomplete because a player's payoff depends on a parameter (his *type*) that only he observes; Harsanyi's device makes Nature draw the types first, which turns the game into one of imperfect information.

(c) Solution. **Static Bayesian game.** $G = \{A_1, \ldots, A_n; T_1, \ldots, T_n; p_1, \ldots, p_n; u_1, \ldots, u_n\}$: action spaces A_i; type spaces T_i; beliefs $p_i(t_{-i} \mid t_i)$ (from a common prior, so that p_i is the conditional of one joint distribution on T); payoffs $u_i(a_1, \ldots, a_n; t_i)$. Timing: (1) Nature draws $t = (t_1, \ldots, t_n)$ and tells player i only t_i; (2) players choose actions $a_i \in A_i$ simultaneously; (3) payoffs u_i are realized.

Pure strategy. A function $s_i : T_i \to A_i$: it says which action each type of i takes. The strategy space has $|A_i|^{|T_i|}$ elements, and it is what a player with $|T_i|$ types is "really" choosing, even though only one type is realized.

Pure-strategy BNE. A profile $s^* = (s_1^*, \ldots, s_n^*)$ such that for every player i and every type $t_i \in T_i$, the action $s_i^*(t_i)$ solves

$$\max_{a_i \in A_i} \sum_{t_{-i} \in T_{-i}} u_i\big(s_1^*(t_1), \ldots, a_i, \ldots, s_n^*(t_n); t_i\big)\, p_i(t_{-i} \mid t_i).$$

Each type best-responds to the others' *strategies* (functions of their types) averaged over his beliefs. Because every type has positive probability, this interim condition is equivalent to Nash equilibrium of the Bayesian normal form, where player i picks a whole function s_i to maximize ex-ante expected payoff (Tadelis p. 251; Gibbons p. 150).

(e) Verification. `check_E1.py` builds the Bayesian normal form of the game in item 2 and checks that its Nash equilibria are exactly the profiles in which every type passes the interim best-reply test, and that the strategy count is $|A_i|^{|T_i|}$ ($2^2 = 4$ there). Gibbons gives no answer.

Answer. A static Bayesian game is $\{A_i; T_i; p_i; u_i\}$ with Nature drawing types first; a pure strategy is a map $s_i : T_i \to A_i$; a BNE is a profile in which every type of every player maximizes expected payoff given his belief $p_i(\cdot \mid t_i)$ and the others' strategies.

Item 2 (item 2 of the combined list)**: Muñoz-Garcia Ch. 7, Exercise 1** Foundations · difficulty 1/5

Source: Muñoz-Garcia Ch. 7, Exercise 1, page 318–319. The statement is reproduced verbatim in the list (item 2); it is not repeated here.

(a) What is asked. (a) The pure Nash equilibria of the matrix in Fig. 7.1 when $x = 15$ and when $x = 4$ is common knowledge. (b) The BNE when only player 1 observes x (prior $\frac{1}{2}, \frac{1}{2}$): strategy spaces, Bayesian normal form, best-response underlining.

(b) Method. (a) Underline best replies in each 2×2 matrix. (b) Player 1 has two types ($x = 15$, $x = 4$) and two actions, so four strategies $s_1 = (a^{15}, a^4)$; player 2 has one type and two actions. Expected payoffs: average the two realizations of x with weight $\frac{1}{2}$ (Tadelis §12.1; Gibbons §3.1.B).

(c) Solution. (a) Complete information. Player 2's payoffs do not depend on x: $BR_2(A) = C$ ($18 > 7$) and $BR_2(B) = D$ ($11 > 0$). For player 1, $BR_1(D) = B$ ($8 > 6$) for every x; against C he compares x with 8:

- $x = 15$: $BR_1(C) = A$ ($15 > 8$). Nash equilibria (A, C) and (B, D).
- $x = 4$: $BR_1(C) = B$ ($8 > 4$). The only Nash equilibrium is (B, D).

(b) Incomplete information. $S_1 = \{A^{15}A^4, A^{15}B^4, B^{15}A^4, B^{15}B^4\}$ (superscript = value of x observed), $S_2 = \{C, D\}$. The Bayesian normal form has $EU_1 = \frac{1}{2}u_1(\cdot; 15) + \frac{1}{2}u_1(\cdot; 4)$ and EU_2 the average over the realized x of player 2's payoff; e.g. $EU_1(A^{15}B^4, C) = \frac{1}{2}\cdot 15 + \frac{1}{2}\cdot 8 = 11.5$ and $EU_2(A^{15}B^4, C) = \frac{1}{2}\cdot 18 + \frac{1}{2}\cdot 0 = 9$. Best replies underlined:

	C	D
$A^{15}A^4$	9.5, $\underline{18}$	6, 7
$A^{15}B^4$	$\underline{11.5}$, $\underline{9}$	7, $\underline{9}$
$B^{15}A^4$	6, $\underline{9}$	7, $\underline{9}$
$B^{15}B^4$	8, 0	$\underline{8}$, $\underline{11}$

Player 1 best-replies $A^{15}B^4$ to C (11.5) and $B^{15}B^4$ to D (8). Player 2 is indifferent between C and D (9 vs. 9) against $A^{15}B^4$ and against $B^{15}A^4$, and strictly prefers C against $A^{15}A^4$ and D against $B^{15}B^4$. Cells where both payoffs are underlined: $(A^{15}B^4, C)$ and $(B^{15}B^4, D)$.

Interim check. In $(A^{15}B^4, C)$: type 15 plays A ($15 > 8$ against C), type 4 plays B ($8 > 4$), and player 2 earns 9 from either action. In $(B^{15}B^4, D)$: both types of player 1 play B ($8 > 6$ against D) and player 2 earns $11 > 0$ from D.

(d) Interpretation. The first BNE reproduces the informative case: player 1's action reveals his type, and player 2, who cannot condition on it, happens to be indifferent. The second is the complete-information equilibrium (B, D) that survives for both values of x.

(e) Verification. `check_E1.py` recomputes the eight cells and compares them with the book's Fig. 7.5 ($9.5, 18$; $6, 7$; $11.5, 9$; $7, 9$; $6, 9$; $7, 9$; $8, 0$; $8, 11$), and enumerates all 4×2 profiles: exactly two BNE.

Answer. (a) $x = 15$: $(A, C), (B, D)$; $x = 4$: (B, D). (b) BNE: $(A^{15}B^4, C)$ and $(B^{15}B^4, D)$.

Item 3 (item 3 of the combined list): Gibbons Problem 3.4 Foundations · difficulty 2/5

Source: Gibbons Problem 3.4, page 169–170. The statement is reproduced verbatim in the list (item 3); it is not repeated here.

(a) What is asked. All pure-strategy BNE when Nature picks Game 1 or Game 2 with probability $\frac{1}{2}$, player 1 observes which, player 2 does not.

(b) Method. Player 1's strategy is a pair $(s_1(G_1), s_1(G_2)) \in \{T, B\}^2$; player 2 chooses L or R with belief $\frac{1}{2}$ on each game. Find the type-by-type best replies of player 1, then player 2's expected payoffs from L and R (Gibbons §3.1.B, pp. 149–151).

(c) Solution. Player 1 (each type). In Game 1: against L, T (payoff 1) beats B (0); against R he is indifferent (0 either way). In Game 2: against L he is indifferent $(0, 0)$; against R, B (2) beats T (0).

Player 2. Given s_1, with $\mathbf{1}[\cdot]$ an indicator,

$$EU_2(L) = \tfrac{1}{2}\,\mathbf{1}[s_1(G_1) = T] \cdot 1 + \tfrac{1}{2} \cdot 0, \qquad EU_2(R) = \tfrac{1}{2} \cdot 0 + \tfrac{1}{2}\,\mathbf{1}[s_1(G_2) = B] \cdot 2.$$

So L is a best reply iff $\frac{1}{2}\mathbf{1}[s_1(G_1) = T] \geq \mathbf{1}[s_1(G_2) = B]$, and R iff the reverse.

- *Player 2 plays L.* Player 1 must play T in G_1; in G_2 any action is a best reply to L. L is optimal for 2 only if $\frac{1}{2} \geq \mathbf{1}[s_1(G_2) = B]$, i.e. $s_1(G_2) = T$. Profile (TT, L): $EU_2(L) = \frac{1}{2} > 0 = EU_2(R)$. ✓
- *Player 2 plays R.* Player 1 must play B in G_2; in G_1 any action is a best reply. R is optimal for 2 iff $\mathbf{1}[s_1(G_2) = B] = 1 \geq \frac{1}{2}\mathbf{1}[s_1(G_1) = T]$, which holds. Profiles (TB, R) and (BB, R). ✓

The remaining profiles fail: BT is never a best reply of player 1 (against L it plays B in G_1, strictly worse than T; against R it plays T in G_2, strictly worse than B), and (TB, L) gives $EU_2(L) = \frac{1}{2} < 1 = EU_2(R)$.

(d) Interpretation. Two equilibria are payoff-dominant in different games: (TT, L) pays $(1, 1)$ in G_1 and $(0, 0)$ in G_2, (BB, R) pays $(0, 0)$ in G_1 and $(2, 2)$ in G_2. Player 2's ignorance creates a coordination problem.

(e) Verification. `check_E1.py` enumerates the 4×2 Bayesian normal form: exactly three pure BNE, $\{(TT, L), (TB, R), (BB, R)\}$. Gibbons gives no answer.

Answer. Three pure-strategy BNE: (TT, L), (TB, R), (BB, R), writing player 1's strategy as (action in Game 1, action in Game 2).

Item 4 (item 4 of the combined list): Tadelis Exercise 12.1 Foundations · difficulty 2/5

Source: Tadelis Exercise 12.1, page 266. The statement is reproduced verbatim in the list (item 4); it is not repeated here.

(a) What is asked. Re-solve the chicken game of §12.2.1 for $(R, H, L) = (8, 16, 0)$ and for $(8, 0, 16)$, and say whether equilibrium analysis supports the claim that a society of lenient parents makes teenagers behave worse.

(b) Method. Each kid has type $k \in \{L, H\}$ (cost of an accident), independent and equiprobable. Payoffs of i: swerve $(C) = 0$; drive (D) against C gives R; D against D gives $R/2 - k$. Strategy xy: x if the kid's parents are lenient (L), y if harsh (H); build the 4×4 Bayesian normal form and underline best replies (Tadelis pp. 252–255).

(c) Solution. **Case** $R = 8, H = 16, L = 0$ (book's matrix, rows CC, CD, DC, DD, columns cc, cd, dc, dd):

	cc	cd	dc	dd
CC	$0, 0$	$0, 4$	$0, 4$	$0, 8$
CD	$4, 0$	$-1, -1$	$-1, 3$	$-6, 2$
DC	$4, 0$	$3, -1$	$\underline{3}, \underline{3}$	$2, 2$
DD	$8, 0$	$2, -6$	$2, 2$	$-4, -4$

The unique pure BNE is (DC, dc): lenient kids $(k = 0)$ drive, harsh kids $(k = 16)$ swerve. *Preacher's claim:* in a society where *all* parents are lenient, both kids have $k = 0$ and D is strictly dominant ($4 > 0$ against D, $8 > 0$ against C): everybody drives and the accident probability is 1, against $\frac{1}{4}$ in the mixed society (only lenient–lenient pairs crash). The analysis supports the claim.

Case $R = 8, H = 0, L = 16$**.** Now lenient parents impose the *high* cost. Matrix:

	cc	cd	dc	dd
CC	$0, 0$	$0, 4$	$0, 4$	$0, 8$
CD	$4, 0$	$\underline{3}, \underline{3}$	$3, -1$	$2, 2$
DC	$4, 0$	$-1, 3$	$-1, -1$	$-6, 2$
DD	$8, 0$	$2, 2$	$2, -6$	$-4, -4$

The unique pure BNE is (CD, cd): lenient kids swerve, harsh $(k = 0)$ kids drive (D is dominant for them). Check for a lenient kid: the rival drives with probability $\frac{1}{2}$ at least, so D yields $\frac{1}{2} \cdot 8 + \frac{1}{2}(4 - 16) = -2 < 0$. The prediction reverses: it is the *harsh*-parent kids who are aggressive. A society of all-lenient parents ($k = 16$ for both) is an ordinary chicken game with a symmetric mixed equilibrium in which each drives with probability q solving $8(1 - q) + q(4 - 16) = 0$, $q = \frac{2}{5}$, so accidents have probability $\frac{4}{25} < \frac{1}{4}$, and the pure asymmetric equilibria have none. The claim is not supported.

(d) Interpretation. The conclusion is driven by the payoff story, not by the equilibrium concept: the same model supports or reverses the preacher depending on whether leniency lowers or raises the perceived cost of an accident. This is exactly the book's own caveat after the first matrix.

(e) Verification. `check_E1.py` computes both 4×4 Bayesian normal forms from the types (first case matches the printed entries $(CD, dd) = (-6, 2)$ and $(DC, dc) = (3, 3)$) and enumerates all pure BNE: one each; it also checks $q = \frac{2}{5}$. Tadelis gives no answer.

Answer. $(R, H, L) = (8, 16, 0)$: unique BNE (DC, dc); the all-lenient society drives for sure, so the claim is supported. $(8, 0, 16)$: unique BNE (CD, cd) (harsh kids drive); the all-lenient society is a chicken game with accident probability $\leq \frac{4}{25}$; the claim is not supported.

Item 5 (item 5 of the combined list): Tadelis Exercise 13.1 Foundations · difficulty 1/5

Source: Tadelis Exercise 13.1, page 285. The statement is reproduced verbatim in the list (item 5); it is not repeated here.

(a) What is asked. Show that bidding one's valuation weakly dominates bidding above it in a second-price sealed-bid auction.

(b) Method. Fix the highest rival bid r (any number) and compare payoffs state by state (Tadelis §13.1.1, pp. 271–273). Bidder i has value v and bids b; if he wins he pays r.

(c) Solution. Let $b > v$. Payoff from bidding v: $v - r$ if $r < v$, 0 if $r > v$ (a tie at $r = v$ pays $v - r = 0$). Payoff from bidding b: $v - r$ if $r < b$, 0 if $r > b$.

rival bid r	bid v	bid $b > v$
$r < v$	$v - r > 0$	$v - r$ (same)
$v \le r < b$	0	$v - r \le 0$ (wins and pays $r \ge v$)
$r > b$	0	0 (same)

In no state does b do better than v; in the middle state ($v < r < b$) it does strictly worse ($v - r < 0$). So $b = v$ weakly dominates every $b > v$. (Tie at $r = b$ pays $v - r \le 0$ in expectation: no gain either.)

(d) Interpretation. Overbidding only adds the states where the bidder wins at a price above his value; the price is the rival's bid, not his own, so he gains nothing in the states where he already won. Underbidding is dominated by the symmetric argument (Muñoz-Garcia item 6).

(e) Verification. `check_E1.py` checks on a grid of 13 values, 13 bids and 13 rival bids (fair coin at ties) that $b = v$ is never worse than any b, and strictly better for some r whenever $b \neq v$. Tadelis gives no answer.

Answer. For every highest rival bid r, bidding v earns at least what any $b > v$ earns, and strictly more when $v < r < b$. Hence bidding the valuation weakly dominates bidding above it.

Item 6 (item 6 of the combined list): Muñoz-Garcia Ch. 8, Exercise 4 Foundations · difficulty 2/5

Source: Muñoz-Garcia Ch. 8, Exercise 4, page 362. The statement is reproduced verbatim in the list (item 6); it is not repeated here.

(a) What is asked. (a) Bidding one's value is weakly dominant in the second-price auction with N bidders. (b) Whether correlated values change optimal bids. (c) Whether N or risk aversion changes them.

(b) Method. Let $r = \max_{j \neq i} b_j$. A weakly dominant strategy is optimal against *every* profile of rivals, hence against every belief about them (Tadelis §13.1.1, pp. 271–273).

(c) Solution. **(a)** Compare $b_i = v_i$ with $b_i \neq v_i$ for each r. If $b_i > v_i$: for $r < v_i$ both win at price r; for $v_i < r < b_i$ the deviation wins and earns $v_i - r < 0$ against 0; for $r > b_i$ both lose. If $b_i < v_i$: for $r < b_i$ both win at r; for $b_i < r < v_i$ the deviation loses and earns 0 against $v_i - r > 0$; for $r > v_i$ both lose. A tie at $r = b_i$ pays no more than $v_i - r$ times the winning probability, never more than bidding v_i. So $b_i = v_i$ is never worse and is strictly better in one state: weak dominance.

(b) No. Correlation changes i's belief about r, but dominance holds state by state, so it holds under any joint distribution of (v_{-i}, b_{-i}): positive or negative correlation, or any N-dimensional dependence, leave $b_i = v_i$ optimal.

(c) No, to both. N enters only through the distribution of r, which is irrelevant for a dominant strategy. Risk aversion: for any strictly increasing utility u with $u(0) = 0$, the state-by-state ranking of $u(v_i - r)$ vs. $u(0)$ is the same as for money, since $v_i - r \gtrless 0$ decides each comparison. The bidding function is the 45° line $b_i(v_i) = v_i$ whatever N and preferences.

(d) Interpretation. Contrast with the first-price auction (items 7, 15): there the bid trades off winning probability against the price paid, so it depends on N, the value distribution and risk attitude. In the second-price auction the price paid does not depend on one's own bid.

(e) Verification. `check_E1.py`: (i) pointwise dominance on a 13^3 grid for three utilities (linear, concave, convex with $u(0) = 0$); (ii) for $N = 2, 3, 5$ and 200 arbitrary joint laws of rival bids, expected payoff from $b = v$ is at least that from every grid bid. M&R's printed answer is the same on all three parts.

Answer. (a) $b_i = v_i$ is weakly dominant. (b) No: dominance is independent of how rivals' values are distributed or correlated. (c) No: neither N nor risk aversion affects the optimal bid.

Item 7 (item 7 of the combined list)**: Muñoz-Garcia Ch. 8, Exercise 1** Foundations · difficulty 2/5

Source: Muñoz-Garcia Ch. 8, Exercise 1, page 350. The statement is reproduced verbatim in the list (item 7); it is not repeated here.

(a) What is asked. First-price auction, two bidders, $v_i \sim U[0,1]$ independent: (a) set up bidder i's expected-utility problem; (b) find the symmetric BNE bidding function.

(b) Method. Look for $b_i(v_i) = \alpha v_i$ with $\alpha \in (0,1)$ (Gibbons §3.2.B, pp. 155–158; Tadelis §13.1.3, pp. 276–279). A bid x wins iff the rival's bid is below x.

(c) Solution. **(a)** If i bids x and the rival uses $b_j = \alpha v_j$, then $\Pr(\text{win}) = \Pr(\alpha v_j < x) = \Pr(v_j < x/\alpha) = x/\alpha$ (for $x \le \alpha$). So

$$EU_i(x \mid v_i) = \underbrace{\frac{x}{\alpha}}_{\Pr(\text{win})} (v_i - x) \; + \; \left(1 - \frac{x}{\alpha}\right) \cdot 0 = \frac{x\,(v_i - x)}{\alpha}.$$

Interpretation: bidding more raises the probability of winning but lowers the margin $v_i - x$ earned on a win; bids above v_i give a negative margin, so $\alpha < 1$ (bid shading).

(b) First-order condition: $\dfrac{\partial EU_i}{\partial x} = \dfrac{v_i - 2x}{\alpha} = 0 \Rightarrow x = \dfrac{v_i}{2}$; second derivative $-2/\alpha < 0$. Consistency with the assumed form $x = \alpha v_i$ requires $\alpha = \frac{1}{2}$. Hence

$$b_i^*(v_i) = \tfrac{1}{2}\, v_i, \qquad EU_i^*(v_i) = \frac{(v_i/2)(v_i/2)}{1/2} = \frac{v_i^2}{2}.$$

Example: $v_i = \frac{2}{3}$ gives the bid $\frac{1}{3}$, as in the book.

Revenue. The winner's bid is $\frac{1}{2}\max(v_1, v_2)$; $E[\max] = \frac{2}{3}$, so expected revenue is $\frac{1}{3}$, equal to the second-price revenue $E[\min(v_1, v_2)] = \frac{1}{3}$ (revenue equivalence).

(d) Interpretation. Half of the value is "paid" for the competition of a single rival; with n bidders the shading falls to v/n (item 15).

(e) Verification. `check_E1.py`: sympy solves the FOC ($x = v/2$, then $\alpha = \frac{1}{2}$); scipy maximizes $\min(2x, 1)(v - x)$ numerically for five values of v and returns $v/2$ with value $v^2/2$; sympy integrates both revenues to $\frac{1}{3}$. Matches M&R's printed $x(v_i) = \frac{1}{2}v_i$.

Answer. (a) $EU_i(x \mid v_i) = \frac{x}{\alpha}(v_i - x)$ against the rival's $b_j = \alpha v_j$. (b) $b_i^*(v_i) = v_i/2$ for both bidders.

Item 8 (item 8 of the combined list)**: Tadelis Exercise 13.2** Foundations · difficulty 2/5

Source: Tadelis Exercise 13.2, page 285. The statement is reproduced verbatim in the list (item 8); it is not repeated here.

(a) What is asked. N bidders in a second-price auction with complete information: (a) is bidding one's value still weakly dominant? (b) are there other Nash equilibria?

(b) Method. (a) The dominance argument of item 5 never used the others' information. (b) Look for profiles where the winner is not the highest-value bidder, using that the winner pays the second-highest bid (Tadelis §13.1.1).

(c) Solution. **(a) Yes.** The state-by-state comparison with the highest rival bid r holds for every r whatever the bidders know, so $b_i = v_i$ is weakly dominant (items 5–6); in particular $(v_1, \ldots, v_N)$ is a Nash equilibrium. With $v_1 > v_2 \geq \ldots$, bidder 1 wins and pays v_2.

(b) Yes, many. Take any bidder k and let $b_k = \max_{j \neq k} v_j$, $b_j = 0$ for $j \neq k$ ($N \geq 2$). Bidder k wins and pays 0, earning $v_k \geq 0$; he cannot do better and has no reason to deviate. Any other bidder j can win only by bidding $> b_k$ and then pays $b_k = \max_{l \neq k} v_l \geq v_j$, earning $v_j - b_k \leq 0$. So no one deviates: a Nash equilibrium in which bidder k wins even if his value is lowest. Example, $v = (3, 2, 1)$: the profile $(0, 3, 0)$ is an equilibrium with the value-2 bidder winning at price 0; so is $(0, 0, 3)$ with the value-1 bidder winning.

(d) Interpretation. These equilibria use weakly dominated strategies ($b_k > v_k$ is weakly dominated by $b_k = v_k$) and rest on the other bidders' staying out; under complete information nothing in Nash equilibrium rules them out. Eliminating weakly dominated strategies selects the truthful equilibrium, as in item 5.

(e) Verification. `check_E1.py` enumerates all profiles on the integer bid grid $\{0, \ldots, 4\}^3$ for $v = (3, 2, 1)$ (fair coin at ties): the truthful profile and $(0, 3, 0)$, $(0, 0, 3)$ are Nash; 55 pure equilibria in all, in which every bidder wins in some equilibrium; dominance of $b_i = v_i$ is verified against every pair of rival bids. Tadelis gives no answer.

Answer. (a) Yes, bidding the value is still weakly dominant. (b) Yes: e.g. one bidder bids the highest rival value and everyone else bids 0; that bidder wins at price 0, whoever he is.

Item 9 (item 9 of the combined list): Tadelis Exercise 12.2 Exam · difficulty 3/5

Source: Tadelis Exercise 12.2, page 266. The statement is reproduced verbatim in the list (item 9); it is not repeated here.

(a) What is asked. The BNE of the Cournot duopoly $p = a - b(q_1 + q_2)$ in which each firm independently has cost c_L with probability μ and $c_H > c_L$ with probability $1 - \mu$, privately known.

(b) Method. Types $T_i = \{c_L, c_H\}$, actions $q_i \geq 0$, beliefs $p_i(c_j \mid c_i) = \mu$ for c_L (independent types). A strategy is a pair $(q_i(c_L), q_i(c_H))$. Write the FOC of each type, given the rival's *average* output $\bar{q}_j = \mu q_j(c_L) + (1-\mu) q_j(c_H)$ (Gibbons §3.1.A, pp. 144–149; Tadelis §12.2.2). Assume interior solutions.

(c) Solution. A firm of type c maximizes $E[(a - b(q + q_j) - c)\, q] = (a - c - b\bar{q}_j - bq)\, q$: $a - c - b\bar{q}_j - 2bq = 0$, i.e.

$$q_i(c) = \frac{a - c - b\bar{q}_j}{2b}, \qquad \text{concave since } -2b < 0.$$

The two types of each firm give the system ($\bar{c} = \mu c_L + (1-\mu) c_H$)

$$q_L = \frac{a - c_L - b\bar{q}}{2b}, \quad q_H = \frac{a - c_H - b\bar{q}}{2b}, \quad \bar{q} = \mu q_L + (1-\mu) q_H.$$

Averaging the first two: $2b\bar{q} = a - \bar{c} - b\bar{q}$, so $\bar{q} = \dfrac{a - \bar{c}}{3b}$. Substituting,

$$q_L^* = \frac{2a - 3c_L + \bar{c}}{6b} = \frac{a - c_L}{3b} + \frac{(1-\mu)(c_H - c_L)}{6b}, \qquad q_H^* = \frac{2a - 3c_H + \bar{c}}{6b} = \frac{a - c_H}{3b} - \frac{\mu(c_H - c_L)}{6b}.$$

The linear system for the four unknowns (two firms, two types) has a unique solution, so the equilibrium is symmetric and unique: both firms use (q_L^*, q_H^*).

(d) Interpretation. Complete-information Cournot gives $(a - c)/(3b)$ for a firm facing a rival with the same cost. Here the low-cost type produces *more* than that when the rival may be high-cost ($1 - \mu > 0$) and the high-cost type produces *less* when the rival may be low-cost: each type adjusts to the rival's expected output.

(e) Verification. `check_E2.py`: sympy solves both the 2-equation and the full 4-equation system and confirms the formulas and $\bar{q} = (a - \bar{c})/(3b)$; scipy maximizes expected profit against the equilibrium rival for $(a, b, c_L, c_H, \mu) = (10, 1, 1, 3, \frac{2}{5})$ and returns q_L^*, q_H^*. Tadelis gives no answer.

Answer. $q^*(c_L) = \dfrac{a - c_L}{3b} + \dfrac{(1-\mu)(c_H - c_L)}{6b}$ and $q^*(c_H) = \dfrac{a - c_H}{3b} - \dfrac{\mu(c_H - c_L)}{6b}$ for both firms (interior case).

Item 10 (item 10 of the combined list): Gibbons Problem 3.3 Exam · difficulty 3/5

Source: Gibbons Problem 3.3, page 169. The statement is reproduced verbatim in the list (item 10); it is not repeated here.

(a) What is asked. Spell out the Bayesian game (actions, types, beliefs, payoffs, strategies) of Bertrand duopoly with demand $q_i = a - p_i - b_i p_j$, $b_i \in \{b_H, b_L\}$ private, write the symmetric pure BNE conditions and solve.

(b) Method. Gibbons §3.1.A (pp. 144–149). The type is the demand sensitivity b_i; firm i's profit is $\pi_i = p_i\,(a - p_i - b_i p_j)$ (zero cost).

(c) Solution. **The game.** Players $i = 1, 2$. Actions $A_i = [0, \infty)$ (prices). Types $T_i = \{b_H, b_L\}$. Beliefs $p_i(b_j \mid b_i) = \theta$ for $b_j = b_H$ and $1 - \theta$ for b_L, independent of b_i. Payoffs $u_i(p_i, p_j; b_i) = p_i(a - p_i - b_i p_j)$. Strategies $p_i : T_i \to [0, \infty)$, i.e. a pair $(p_i(b_H), p_i(b_L))$.

BNE conditions. A symmetric pure profile (p_H, p_L) (same function for both firms) is a BNE iff for each $b \in \{b_H, b_L\}$

$$p_b \in \arg\max_p \;\theta\, p\,(a - p - b\,p_H) + (1-\theta)\, p\,(a - p - b\,p_L) = \arg\max_p \;p\,(a - p - b\,\bar{p}),$$

where $\bar{p} = \theta p_H + (1-\theta) p_L$ (the rival's expected price; profit is linear in p_j).

Solution. FOC $a - 2p - b\bar{p} = 0$ (strictly concave: $-2 < 0$), so $p_b = (a - b\bar{p})/2$. Averaging over types with weights $\theta, 1-\theta$: $\bar{p} = \dfrac{a - \bar{b}\bar{p}}{2}$, where $\bar{b} = \theta b_H + (1-\theta) b_L$, so $\bar{p} = \dfrac{a}{2 + \bar{b}}$ and

$$p_H^* = \frac{a}{2}\Big(1 - \frac{b_H}{2 + \bar{b}}\Big) = \frac{a\,(2 + \bar{b} - b_H)}{2\,(2 + \bar{b})}, \qquad p_L^* = \frac{a\,(2 + \bar{b} - b_L)}{2\,(2 + \bar{b})}.$$

Since $b_H > b_L$, $p_H^* < p_L^*$: the type whose demand reacts more to the rival's price prices lower. Positivity requires $2 + \bar{b} > b_H$, true e.g. for $b_H < 2$.

(d) Interpretation. With $b_H = b_L = b$ the price is $a/(2+b)$, the standard differentiated-products Bertrand price. Here the rival's type is unknown, so each firm prices against the *expected* rival price.

(e) Verification. `check_E2.py`: sympy solves the two FOCs for (p_H, p_L) and matches the formulas and $\bar{p} = a/(2 + \bar{b})$; scipy maximizes profit for $(a, b_H, b_L, \theta) = (10, \frac{3}{2}, \frac{1}{2}, \frac{1}{3})$ against the equilibrium rival and returns p_H^*, p_L^*. Gibbons gives no answer.

Answer. Symmetric BNE: $p^*(b) = \dfrac{a\,(2 + \bar{b} - b)}{2\,(2 + \bar{b})}$ with $\bar{b} = \theta b_H + (1-\theta) b_L$, for $b \in \{b_H, b_L\}$.

Item 11 (item 11 of the combined list): Tadelis Exercise 12.3 Exam · difficulty 3/5

Source: Tadelis Exercise 12.3, page 266–267. The statement is reproduced verbatim in the list (item 11); it is not repeated here.

(a) What is asked. All pure-strategy BNE of the armed-conflict game for (a) $m = 3, w = 2, s = 1$ and (b) $m = 3, w = 4, s = 2$.

(b) Method. Types S or W (equiprobable, independent), actions A or N. An army's payoff from N is 0. From A: m if the rival plays N; if both attack, $m - c$ when i is strong and the rival weak, and $-c$ otherwise, where $c = s$ (strong) or w (weak). A strategy is (x_S, x_W), the action of the strong and of the weak type. For each type compute the expected payoff of A against each of the four rival strategies and compare with 0.

(c) Solution. Expected payoff of A for type t against a rival strategy (y_S, y_W) is $\frac{1}{2}u(A \mid y_S \text{ vs } S) + \frac{1}{2}u(A \mid y_W \text{ vs } W)$; ($N$ yields 0).

(a) $m = 3, w = 2, s = 1$.

rival (y_S, y_W)	AA	AN	NA	NN
$E[A]$, strong	$\frac{1}{2}$	1	$\frac{5}{2}$	3
$E[A]$, weak	-2	$\frac{1}{2}$	$\frac{1}{2}$	3

The strong type attacks against every rival strategy ($E[A] > 0$). The weak type attacks unless the rival is AA (always attacks). So a rival AA forces (A, N) from the other army, but against AN the weak type wants A: no symmetric pure equilibrium. The BNE are asymmetric:

$$(AA,\ AN) \quad \text{and} \quad (AN,\ AA).$$

Check (AA, AN): army 1 faces AN and attacks with both types ($1 > 0$, $\frac{1}{2} > 0$); army 2 faces AA, so its strong type attacks ($\frac{1}{2} > 0$) and its weak type stays ($-2 < 0$).

(b) $m = 3, w = 4, s = 2$.

rival (y_S, y_W)	AA	AN	NA	NN
$E[A]$, strong	$-\frac{1}{2}$	$\frac{1}{2}$	2	3
$E[A]$, weak	-4	$-\frac{1}{2}$	$-\frac{1}{2}$	3

The weak type attacks only if the rival never attacks ($E[A] = 3$ against NN, negative otherwise); the strong type attacks unless the rival is AA. Profiles:

$$(AN,\ AN), \qquad (AA,\ NN), \qquad (NN,\ AA).$$

Check (AN, AN): strong attacks ($\frac{1}{2} > 0$ against AN), weak stays ($-\frac{1}{2} < 0$). (AA, NN): army 1 attacks with both types because army 2 never does ($3 > 0$), and army 2 stays with both types because army 1 always attacks ($-\frac{1}{2} < 0$ and $-4 < 0$).

(d) Interpretation. (a) The costs are low enough that attacking pays unless the rival is sure to attack. The only equilibria have one army whose weak type attacks and another whose weak type backs off. (b) With high costs of fighting, only the strong type fights; the weak type attacks only against a rival who never does, which supports the lopsided equilibria (AA, NN) and (NN, AA).

(e) Verification. `check_E2.py` computes both Bayesian normal forms from the payoff rules and enumerates all 4×4 pure profiles; it prints the two tables of expected attack payoffs above and asserts exactly the two profiles in (a) and the three in (b). Tadelis gives no answer.

Answer. (a) (AA, AN) and (AN, AA). (b) (AN, AN), (AA, NN) and (NN, AA); each strategy lists the action of the strong type, then of the weak type.

Item 12 (item 12 of the combined list): Tadelis Exercise 12.9 Exam · difficulty 3/5

Source: Tadelis Exercise 12.9, page 269. The statement is reproduced verbatim in the list (item 12); it is not repeated here.

(a) What is asked. (a) Nash equilibria of grab-the-dollar. (b) The pure-strategy BNE when $\theta_i \sim U[-\varepsilon, \varepsilon]$ is added to the payoff of the sole grabber. (c) Its limit as $\varepsilon \to 0$.

(b) Method. (a) Best replies; indifference gives the mixed equilibrium. (b) A player grabs when the expected payoff from grabbing is nonnegative, which is increasing in θ_i: a threshold rule (Tadelis §12.5, pp. 264–266, Harsanyi's purification).

(c) Solution. **(a)** Payoffs: sole grabber 1, non-grabber 0, both grab -1 each. Pure Nash equilibria: (G, D) and (D, G). Mixed: if the rival grabs with probability p, grabbing yields $(1-p)\cdot 1 + p\cdot(-1) = 1-2p$ against 0 from not grabbing; indifference gives $p = \frac{1}{2}$. So the mixed equilibrium has both players grabbing with probability $\frac{1}{2}$.

(b) Let the rival grab with probability p. Player i of type θ_i gets $(1-p)(1+\theta_i) - p$ from G and 0 from D, so he grabs iff

$$\theta_i \;\geq\; \frac{2p-1}{1-p}.$$

Take thresholds t_i (i grabs iff $\theta_i \geq t_i$): the rival grabs with probability $p_j = \dfrac{\varepsilon - t_j}{2\varepsilon}$, hence $1 - p_j = \dfrac{\varepsilon + t_j}{2\varepsilon}$, $2p_j - 1 = -t_j/\varepsilon$ and

$$t_i = \frac{-2\,t_j}{\varepsilon + t_j}, \qquad t_j = \frac{-2\,t_i}{\varepsilon + t_i}.$$

Subtracting the two cross-multiplied equations $t_i(\varepsilon+t_j) = -2t_j$ and $t_j(\varepsilon+t_i) = -2t_i$ gives $(\varepsilon-2)(t_i-t_j) = 0$, so $t_i = t_j = t$ because $\varepsilon < 1 < 2$; then $t(\varepsilon + t) = -2t$, i.e. $t = 0$ or $t = -(2+\varepsilon)$, and only $t = 0$ lies in $[-\varepsilon, \varepsilon]$. The interior pure BNE is: *each player grabs iff* $\theta_i \geq 0$, i.e. with probability $\frac{1}{2}$. (Two further pure BNE sit at the corners: one player always grabs, since $1 + \theta \geq 1 - \varepsilon > 0$ when the other never grabs, and the other never grabs, since grabbing against a sure grabber yields -1. They are the perturbed versions of (G, D) and (D, G).)

(c) In the symmetric BNE, the probability of grabbing is $\frac{1}{2}$ for *every* ε, so as $\varepsilon \to 0$ it converges to the mixed-strategy equilibrium of (a), now interpreted as a pure strategy of a slightly uncertain game: players randomize only in the eyes of an observer who does not know their θ_i.

(d) Interpretation. The mixed equilibrium becomes the limit of pure-strategy BNE, with no player randomizing: this is Harsanyi's purification argument.

(e) Verification. `check_E2.py`: sympy solves the two cross equations and returns only $(t_1, t_2) = (0, 0)$ in $[-\varepsilon, \varepsilon]$; the grab probability at $t = 0$ equals $\frac{1}{2}$; the mixed $p = \frac{1}{2}$ is checked. Tadelis gives no answer.

Answer. (a) (G, D), (D, G) and the mixed equilibrium with $p = \frac{1}{2}$. (b) Symmetric BNE: grab iff $\theta_i \geq 0$ (probability $\frac{1}{2}$); there are also the two corner pure BNE. (c) The grabbing probability is $\frac{1}{2}$ for all ε, hence converges to the mixed equilibrium of (a).

Item 13 (item 13 of the combined list): Tadelis Exercise 12.5 Exam · difficulty 3/5

Source: Tadelis Exercise 12.5, page 267–268. The statement is reproduced verbatim in the list (item 13); it is not repeated here.

(a) What is asked. (a) For a price p, the prospector's threshold type; (b) the pure BNE, its uniqueness and the expected payoffs.

(b) Method. The prospector knows $x \sim U[0,1]$; the owner does not. This is adverse selection (Tadelis §12.3, pp. 258–261): the types that sell are the worst ones, which the owner must take into account in his expected payoff. Find the owner's expected net value conditional on acceptance.

(c) Solution. **(a)** A prospector of type x who asks p sells iff the price is at least his self-mining value: $p \geq 3x$, i.e. $x \leq x(p) = \min\{p/3,\, 1\}$. The payoff $3x$ increases in x while the price is type-independent, so types below $x(p)$ prefer to sell and types above prefer to mine.

(b) If the owner accepts price $p \in (0,3]$, he buys exactly the types $x \leq p/3$, with $E[x \mid x \leq p/3] = p/6$:

$$E[4x - p \mid x \leq p/3] = 4 \cdot \frac{p}{6} - p = -\frac{p}{3} < 0.$$

For $p \geq 3$ all types sell and $E[4x - p] = 2 - p < 0$. So the owner loses on average at any positive price and rejects it. Accepting $p = 0$ yields 0 (indifference), but then every type $x > 0$ strictly prefers asking a rejected price and mining ($3x > 0$).

Uniqueness. Suppose some price $p > 0$ were accepted in equilibrium; the highest accepted price is asked by every type $x < p/3$ (they gain $p - 3x > 0$ from it), so the owner's conditional payoff is $-p/3 < 0$: accepting is not a best reply, a contradiction. Hence in every pure BNE the owner accepts no positive price that is asked, and trade never occurs (up to the null type $x = 0$).

Payoffs. Each prospector type earns $3x$ (expected $\frac{3}{2}$), the owner earns 0. The market breaks down although trade would create surplus $4x - 3x = x > 0$ for every type.

(d) Interpretation. The owner cannot tell good mines from bad: any price high enough to attract the good ones is above their average value to him. Akerlof's lemons logic, with a continuum of types.

(e) Verification. `check_E2.py`: sympy integrates $E[4x - p \mid x \leq p/3] = -p/3$, $E[x \mid x \leq p/3] = p/6$, the full-sale case $2 - p$ and $E[3x] = \frac{3}{2}$. Tadelis gives no answer.

Answer. (a) $x(p) = \min\{p/3, 1\}$. (b) Unique pure BNE: no trade (the owner rejects every $p > 0$, since $E[4x - p \mid x \leq p/3] = -p/3$); payoffs $3x$ to the prospector, 0 to the owner.

Item 14 (item 14 of the combined list): Tadelis Exercise 12.6 Exam · difficulty 4/5

Source: Tadelis Exercise 12.6, page 268. The statement is reproduced verbatim in the list (item 14); it is not repeated here.

(a) What is asked. (a) Strategies and pure BNE of the farmer–plant game; (b) the social surplus in equilibrium; (c) the effect on surplus of cutting the farming payoff to $\theta/2$.

(b) Method. As in item 13: a farmer of talent $\theta \sim U[0,1]$ asks a wage w; the owner accepts or rejects. Strategies: farmer $w : [0,1] \to [0,\infty)$; owner $\{A, R\}$ as a function of w. The owner's expected net value is computed on the set of types that accept (Tadelis §12.3, pp. 258–261).

(c) Solution. **(a)** A farmer of type θ who asks w takes the job iff $w \geq \theta$ (outside option θ), so types $\theta \leq w$ sell. If the owner accepts $w \in (0,1]$ he employs the types $\theta \leq w$, with $E[\theta \mid \theta \leq w] = w/2$:

$$E\big[\tfrac{3}{2}\theta - w \mid \theta \leq w\big] = \tfrac{3}{4}w - w = -\tfrac{w}{4} < 0; \qquad w \geq 1 : \ \tfrac{3}{4} - w < 0.$$

The owner rejects every positive wage; at $w = 0$ he is indifferent, but farmers $\theta > 0$ prefer to farm. As in item 13, an accepted positive wage would be asked by every type $\theta < w$ and would give the owner a loss, so it cannot occur in a BNE. *BNE:* the owner rejects all $w > 0$ (and may accept $w = 0$); every farmer farms (any rejected ask is a best reply).

(b) Surplus = farmer's plus owner's expected payoffs $= E[\theta] + 0 = \frac{1}{2}$. It is the only possible level: no trade occurs in any BNE (the efficient level would be $E[\frac{3}{2}\theta] = \frac{3}{4}$).

(c) Now the farmer's outside option is $\theta/2$, so types $\theta \leq 2w$ accept a wage w. For $w \leq \frac{1}{2}$: $E[\frac{3}{2}\theta - w \mid \theta \leq 2w] = \frac{3}{2}w - w = \frac{w}{2} > 0$; for $w \geq \frac{1}{2}$ all types sell and $E = \frac{3}{4} - w \geq 0$ iff $w \leq \frac{3}{4}$. So the owner accepts any wage up to $\frac{3}{4}$, and a farmer asks the highest accepted wage, getting $\frac{3}{4} > \theta/2$ for all θ. The BNE with trade at a wage w^* accepted by the owner has surplus

$$S(w^*) = \tfrac{1}{4} + 2w^{*2} \ (w^* \leq \tfrac{1}{2}), \qquad S(w^*) = \tfrac{3}{4} \ (\tfrac{1}{2} \leq w^* \leq \tfrac{3}{4}),$$

and the no-trade BNE has surplus $\frac{1}{4}$. The maximum, $\frac{3}{4}$, is reached when everybody works: with the owner's best reply (accept every wage that gives him a nonnegative net value) the farmer asks $w = \frac{3}{4}$ and trade is efficient. Since $\frac{3}{4} > \frac{1}{2}$, the criterion *supports* the policy-maker: cutting the subsidy turns a market that had broken down into one with full trade. Caveats: no-trade remains a BNE with surplus $\frac{1}{4} < \frac{1}{2}$ (and trade at $w^* < \sqrt{2}/4 \approx 0.35$ gives less than $\frac{1}{2}$); and the farmer types $\theta > \frac{3}{4}$ are worse off than before the policy (they now earn $\frac{3}{4} < \theta$).

(d) Interpretation. With outside option $\theta/2$ the average talent of those willing to work at wage w, $E[\theta \mid \theta \leq 2w] = w$, is high enough to cover the wage ($\frac{3}{2}w > w$), so adverse selection no longer blocks trade: a worse outside option can raise total surplus.

(e) Verification. `check_E2.py`: sympy confirms $-w/4$ in (a); $E[\theta] = \frac{1}{2}$; in (c) $w/2$, $\frac{3}{4} - w$ (zero at $w = \frac{3}{4}$), the surplus function $\frac{1}{4} + 2w^2$ (equal to $\frac{3}{4}$ at $w = \frac{1}{2}$), the no-trade surplus $\frac{1}{4}$ and the break-even $w = \sqrt{2}/4$. Tadelis gives no answer.

Answer. (a) No-trade BNE: the owner rejects every $w > 0$ ($E = -w/4$). (b) Surplus $\frac{1}{2}$. (c) With outside option $\theta/2$ the owner accepts wages up to $\frac{3}{4}$, and the best BNE has all farmers work at $w = \frac{3}{4}$ with surplus $\frac{3}{4} > \frac{1}{2}$: the policy can be advocated on the surplus criterion (not every BNE improves, and types $\theta > \frac{3}{4}$ lose).

Item 15 (item 15 of the combined list): Gibbons Problem 3.6 Exam · difficulty 3/5

Source: Gibbons Problem 3.6, page 170. The statement is reproduced verbatim in the list (item 15); it is not repeated here.

(a) What is asked. Show that with n bidders with values i.i.d. $U[0,1]$ in a first-price sealed-bid auction, bidding $\frac{n-1}{n}v_i$ is a symmetric BNE.

(b) Method. Verification, not derivation: take the rivals' strategy $b_j = \frac{n-1}{n}v_j$ as given and show that it is a best reply for every v_i (Gibbons §3.2.B, pp. 155–158).

(c) Solution. Let $k = \frac{n-1}{n}$. A bid $x \le k$ beats rival j iff $kv_j < x$, i.e. $v_j < x/k$, with probability x/k; it beats all $n-1$ independent rivals with probability $(x/k)^{n-1}$. Hence

$$EU_i(x \mid v_i) = \left(\frac{x}{k}\right)^{n-1}(v_i - x), \qquad 0 \le x \le k.$$

FOC: $(n-1)\dfrac{x^{n-2}}{k^{n-1}}(v_i - x) - \dfrac{x^{n-1}}{k^{n-1}} = 0$, i.e. $(n-1)(v_i - x) = x$, so

$$x = \frac{n-1}{n}\,v_i = kv_i.$$

This is the unique interior critical point and a maximum: on $[0,k]$, $EU_i = c\,x^{n-1}(v_i - x)$ with $c > 0$ is 0 at $x = 0$, increasing up to $x = kv_i$ and decreasing afterwards. Bids $x > k$ win with probability 1 and earn $v_i - x$, decreasing in x and continuous with $EU_i(k)$, so they do no better than $x = k$, which is already beaten by kv_i (as $kv_i \le k$, with equality only at $v_i = 1$). So $b_i(v_i) = kv_i$ is a best reply to the other $n-1$ bidders using it: a symmetric BNE.

(d) Interpretation. Shading is v_i/n: it vanishes as $n \to \infty$. The expected revenue is $E[\max v] \cdot k = \frac{n}{n+1} \cdot \frac{n-1}{n} = \frac{n-1}{n+1}$, the expected second-highest value, i.e. the second-price revenue (revenue equivalence); $n = 2$ gives item 7's $\frac{1}{3}$.

(e) Verification. `check_E2.py`: for $n = 2, \ldots, 7$ sympy solves the FOC and returns $(n-1)v/n$; scipy maximizes EU_i (with the winning-for-sure branch for $x > k$) at $v = 0.7$ and returns $(n-1)\cdot 0.7/n$; sympy integrates the two revenues to $(n-1)/(n+1)$. Gibbons gives no answer.

Answer. With the rivals bidding $\frac{n-1}{n}v_j$, $EU_i = (\frac{nx}{n-1})^{n-1}(v_i - x)$ is maximized at $x = \frac{n-1}{n}v_i$: the profile is a symmetric BNE.

Item 16 (item 16 of the combined list): Tadelis Exercise 13.9 Exam · difficulty 3/5

Source: Tadelis Exercise 13.9, page 287. The statement is reproduced verbatim in the list (item 16); it is not repeated here.

(a) What is asked. Two bidders, first-price auction, values $0.5+\theta_i$ with $\theta_i \sim U[0,1]$: (a) a symmetric BNE of the form $s(\theta_i) = a + b\theta_i$; (b) the expected payoff of type θ_i.

(b) Method. Method of undetermined coefficients (Tadelis §13.1.3, pp. 276–280): let the rival bid $a + b\theta_j$, write the bidder's expected payoff for a bid x, take the FOC and match coefficients.

(c) Solution. **(a)** A bid $x \in [a, a+b]$ beats the rival iff $a + b\theta_j < x$, i.e. $\theta_j < (x-a)/b$, with probability $(x-a)/b$. So

$$EU_i(x \mid \theta_i) = \frac{x-a}{b}\big(0.5 + \theta_i - x\big).$$

FOC: $(0.5+\theta_i - x) - (x-a) = 0$, so $x = \frac{1}{2}(0.5+\theta_i+a)$ (concave in x). Matching with $x = a + b\theta_i$ for all θ_i:

$$b = \tfrac{1}{2}, \qquad a = \tfrac{1}{2}(0.5+a) \;\Rightarrow\; a = \tfrac{1}{2}.$$

So $s^*(\theta_i) = \frac{1}{2} + \frac{1}{2}\theta_i$, with bids in $[\frac{1}{2}, 1]$. Bids below $a = \frac{1}{2}$ win with probability 0, and bids above 1 win surely at a worse price than 1; neither helps.

(b) The probability of winning with $x = s^*(\theta_i)$ is $(x-a)/b = \theta_i$, and the margin is $0.5 + \theta_i - \frac{1}{2} - \frac{1}{2}\theta_i = \frac{1}{2}\theta_i$, so

$$EU_i^*(\theta_i) = \theta_i \cdot \tfrac{1}{2}\theta_i = \frac{\theta_i^2}{2}.$$

(d) Interpretation. In value terms $v_i = 0.5 + \theta_i$, the bid is $v_i/2 + \frac{1}{4}$: half the value plus $\frac{1}{4}$, because the lowest value is 0.5, so no one needs to bid below 0.5.

(e) Verification. `check_E2.py`: sympy gets $x = (0.5 + \theta + a)/2$ and the unique $(a, b) = (\frac{1}{2}, \frac{1}{2})$; scipy maximizes EU_i numerically at $\theta \in \{0.2, 0.5, 0.9, 1\}$ and returns $0.5 + 0.5\theta$; the equilibrium payoff $\theta^2/2$ is confirmed symbolically and numerically. Tadelis gives no answer.

Answer. (a) $s^*(\theta_i) = \frac{1}{2} + \frac{1}{2}\theta_i$ ($a = b = \frac{1}{2}$). (b) $EU_i^*(\theta_i) = \theta_i^2/2$.

Item 17 (item 17 of the combined list): Tadelis Exercise 13.3 Exam · difficulty 3/5

Source: Tadelis Exercise 13.3, page 285–286. The statement is reproduced verbatim in the list (item 17); it is not repeated here.

(a) What is asked. Second-price auction with reserve price r, two buyers, values $\{0, 1, 2\}$ equiprobable and independent: (a) is truthful bidding still weakly dominant? Expected revenue at (b) $r = 0$, (c) $r = 1$; (d) why they differ; (e) the optimal r.

(b) Method. (a) Compare payoffs state by state as in item 5. (b)–(e) Enumerate the nine value pairs (each probability $\frac{1}{9}$) with bids equal to values (Tadelis §13.1.1 and the reserve-price discussion).

(c) Solution. **(a) Yes.** A winner pays $\max\{r, \text{rival bid}\}$ and wins iff his bid is at least r and beats the rival. Value $v < r$: bidding v loses; any winning bid would pay at least $r > v$, a loss. Value $v \geq r$: the situation is that of item 5 with the effective price $\max\{r, \text{rival bid}\}$, so the same table shows no bid beats $b = v$.

(b) $r = 0$**.** Revenue is the second-highest value, $\min(v_1, v_2)$: 0 in 5 pairs, 1 in 3 pairs, 2 in 1 pair:

$$\tfrac{1}{9}(0 \cdot 5 + 1 \cdot 3 + 2 \cdot 1) = \tfrac{5}{9}.$$

(c) $r = 1$**.** Value-0 buyers do not bid. Pairs: $(0, 0)$ no sale (0); one buyer at least 1 and the other 0 (4 pairs): price $r = 1$; both at least 1: price min: $(1, 1), (1, 2), (2, 1) \to 1$ each, $(2, 2) \to 2$. Revenue $= \frac{1}{9}(4 \cdot 1 + 1 + 1 + 1 + 2) = \frac{9}{9} = 1$.

(d) Compared with $r = 0$, the four pairs with exactly one value-0 buyer now yield 1 instead of 0 ($+\frac{4}{9}$): the other buyer used to pay the second-highest bid, 0, and now pays r. The pair $(0, 0)$ yields 0 in both cases and the pairs with both values at least 1 are unaffected (price min $\geq 1 = r$). Nothing is lost, because $r = 1$ is the lowest positive value: it excludes only buyers who would have paid 0.

(e) For $r \in (0, 1]$: revenue $= \frac{5}{9} + \frac{4}{9}r$ (the $(1, 1), (1, 2), (2, 1), (2, 2)$ pairs give 5/9, the four "one buyer" pairs give r). For $r \in (1, 2]$ only value-2 buyers bid: four pairs at price r and $(2, 2)$ at price 2, revenue $= \frac{4r+2}{9}$, increasing in r; $r > 2$ gives 0. The maximum is at $r = 2$ (the highest value) with revenue $\frac{10}{9}$, twice the revenue of $r = 0$ (strictly above $r = 1$'s 1). Conclusion: a reserve price above the seller's value (here 0) raises revenue: the seller gives up sales that would have been at low prices to raise the price paid by the high types; the optimal reserve is above the lowest values.

(d) Interpretation. A reserve price is a monopoly price inside the auction: raising it above 1 also excludes the value-1 buyers, but the extra revenue from value-2 buyers (r instead of min) outweighs the lost sales, because each value-2 buyer is alone in $\frac{4}{9}$ of the pairs.

(e) Verification. `check_E2.py`: enumerates the nine pairs for r on a fine grid and asserts 5/9, 1, 10/9 at $r = 0, 1, 2$, the two linear pieces for $r \in (0, 1]$ and $(1, 2]$, the global maximum 10/9 at $r = 2$; and dominance of $b = v$ against every rival bid for five values of v and four reserves. Tadelis gives no answer.

Answer. (a) Yes. (b) 5/9. (c) 1. (d) Excluding value-0 buyers lets the seller charge 1 in the four pairs with exactly one such buyer, at no loss elsewhere. (e) The optimal reserve is $r = 2$, with revenue 10/9: reserve prices can raise revenue above the no-reserve auction.

Item 18 (item 18 of the combined list): Tadelis Exercise 12.8 Exam · difficulty 4/5

Source: Tadelis Exercise 12.8, page 268–269. The statement is reproduced verbatim in the list (item 18); it is not repeated here.

(a) What is asked. In the jury game of §12.4 (prior $q > \frac{1}{2}$ of guilt, signals with precision $p > \frac{1}{2}$, payoff 1 for a correct verdict): (a) show that CC for both is a BNE of the two-juror (unanimity) game; (b) compare with a single decider; (c) three jurors, majority rule: when is voting by signal a BNE?

(b) Method. A juror's vote matters only when he is *pivotal*; he must condition his belief about guilt on being pivotal (the swing voter's curse, Tadelis pp. 262–264). Strategy xy: x after signal θ_G, y after θ_I. Posteriors by Bayes' rule: $\Pr(G \mid \theta_G) = \dfrac{qp}{qp + (1-q)(1-p)} > q$ and $\Pr(G \mid \theta_I) = \dfrac{q(1-p)}{q(1-p) + (1-q)p} < q$.

(c) Solution. **(a)** If juror j plays CC, i is pivotal in every state and decides alone. He convicts iff $\Pr(G \mid \theta_i) \geq \frac{1}{2}$. After θ_G this holds; after θ_I it holds iff $q(1-p) \geq (1-q)p$, i.e. iff

$$p \leq q.$$

So (CC, CC) is a BNE *if and only if* $p \leq q$; for $p \leq q$ even the innocent-looking signal is too weak to overturn the prior q, so everyone convicts. *This is a point where the book's statement ("both choosing CC is a BNE", p. 264) needs the condition $p \leq q$:* for $p > q$ a juror with θ_I facing a sure convictor is pivotal and strictly prefers A. For $p > q$ the pure BNE are (AA, AA) (nobody can change an acquittal) and the asymmetric (CA, CC), (CC, CA) (one juror always convicts and the other votes by his signal, being pivotal in every state); voting by signal for both, (CA, CA), is not a BNE for any p, because a juror with θ_I who is pivotal knows the other saw θ_G and, with $\Pr(G \mid \theta_I, \theta_G) = \dfrac{q(1-p)p}{q(1-p)p + (1-q)p(1-p)} = q > \frac{1}{2}$, prefers C.

(b) Payoff = probability of a correct verdict. A single decider who votes by signal gets p (the verdict equals his signal, correct with probability p in each state); always convicting gives q; so he does best with $\max\{p, q\}$: by signal iff $p > q$. In the two-juror game: (CC, CC) gives q (when $p \leq q$), (AA, AA) gives $1 - q < q$ and (CA, CC) gives p (when $p > q$). So a single decider is never worse than any equilibrium of the two-juror game ($\max\{p, q\} \geq q$, p, $1 - q$) and is strictly better when $p > q$ than every *symmetric* equilibrium (the only one is (AA, AA), paying $1 - q$); if one reads the book's (CC, CC) as the equilibrium, the single decider's p beats its q when $p > q$. The reason: under unanimity a juror is pivotal only when the other convicts, so the two jurors cannot both use their information (swing voter's curse), while one decider faces no such problem. The only two-juror equilibria that use information, (CA, CC) and (CC, CA), simply make one juror the single decider.

(c) Three jurors, conviction iff at least two votes C. Suppose the others vote by signal (CA). Juror i is pivotal iff the others split, one θ_G and one θ_I (one votes C, one A): probability $2p(1-p)$ in both states of the world, so the split carries no information about G. Then

$$\Pr(G \mid \theta_i = \theta_G, \text{split}) = \frac{qp}{qp + (1-q)(1-p)} > q > \tfrac{1}{2}, \qquad \Pr(G \mid \theta_i = \theta_I, \text{split}) = \frac{q(1-p)}{q(1-p) + (1-q)p}.$$

After θ_G convicting is optimal; after θ_I acquitting is optimal iff the second posterior is at most $\frac{1}{2}$, i.e. $p \geq q$. So (CA, CA, CA) is a BNE iff

$$p \;\geq\; q,$$

the same condition as for a single decider: when the others' split cancels out, a pivotal juror reads only his own signal.

(d) Interpretation. Unanimity makes the pivotal event "the other juror voted C", which erases the juror's own contrary signal; majority of three makes it "the others disagree", which is uninformative. This is why Feddersen–Pesendorfer find that more than two jurors can restore informative voting.

(e) Verification. `check_E2.py` enumerates all pure profiles of the two-juror game for $(q,p) = (\frac{3}{5},\frac{4}{5})$, $(\frac{4}{5},\frac{3}{5})$, $(\frac{3}{5},\frac{3}{5})$: $p>q$: $\{AA,AA;\ CA,CC;\ CC,CA\}$; $p<q$: $\{AA,AA;\ CC,CC\}$; $p=q$: all four. It asserts that (CC,CC) is a BNE iff $p \le q$, that (CA,CA) never is, the single-decider payoffs p and q and, on an 81-point grid of (q,p), that (CA,CA,CA) is a BNE of the three-juror majority game iff $p \ge q$. Tadelis gives no answer.

Answer. (a) (CC,CC) is a BNE iff $p \le q$ (for $p>q$ it fails). (b) A single decider gets $\max\{p,q\}$: at least as much as any equilibrium of the two-juror game, and strictly more than the symmetric ones when $p>q$ (the unanimity rule wastes the jurors' information through the swing voter's curse). (c) Voting by signal is a BNE of the three-juror majority game iff $p \ge q$.

Item 19 (item 19 of the combined list): MWG Exercise 13.B.1 Advanced · difficulty 4/5

Source: MWG Exercise 13.B.1, page 473. The statement is reproduced verbatim in the list (item 19); it is not repeated here.

(a) What is asked. Graph $r(\hat\theta)$, $E[\theta \mid \theta \le \hat\theta]$ and $\hat\theta$ on $[\underline\theta, \bar\theta]$; read off the competitive equilibria with unobservable types and the Pareto optimal allocation; then draw the situations of Figures 13.B.1–13.B.3.

(b) Method. Model of MWG §13.B (pp. 437–443): a worker of type θ produces θ at a firm and $r(\theta)$ at home, and accepts a wage w iff $r(\theta) \le w$. With r increasing, the accepting set at wage $w = r(\hat\theta)$ is $\{\theta \le \hat\theta\}$ and firms break even iff $w = E[\theta \mid \theta \le \hat\theta]$ (Def. 13.B.1). Reading the diagram in the cutoff $\hat\theta$ turns the equilibrium condition into a crossing.

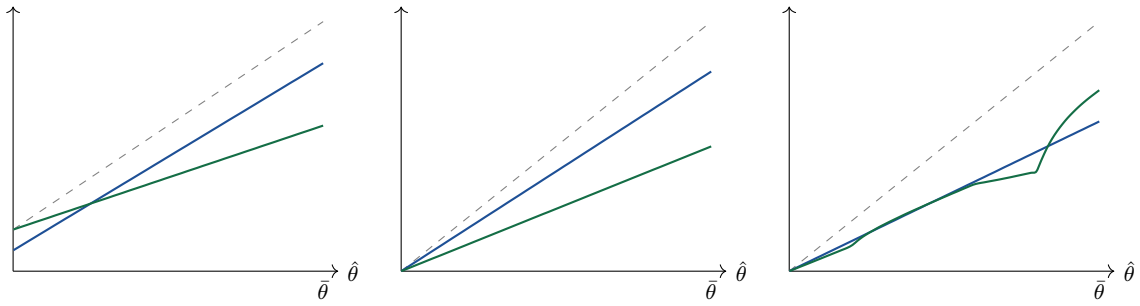
(c) Solution. Competitive equilibria. An interior cutoff $\hat\theta \in (\underline\theta, \bar\theta)$ with wage $w = r(\hat\theta)$ is an equilibrium iff

$$r(\hat\theta) = E[\theta \mid \theta \le \hat\theta],$$

a point where the curve $r(\hat\theta)$ crosses the curve $E[\theta \mid \theta \le \hat\theta]$. Two corners: $\hat\theta = \bar\theta$ (everybody works, wage $E[\theta]$) is an equilibrium iff $r(\bar\theta) \le E[\theta]$; $\hat\theta = \underline\theta$ (only the lowest type, a null set, wage $\underline\theta$) is an equilibrium iff $r(\underline\theta) = \underline\theta$ (the curves meet at the left end). The curve $E[\theta \mid \theta \le \hat\theta]$ is increasing (if $f > 0$), starts at $\underline\theta$, ends at $E[\theta]$ and lies below the 45° line $\hat\theta$.

Pareto optimal allocation. Workers with $r(\theta) \le \theta$ work at a firm: the set of $\hat\theta$ where the curve r lies on or below the 45° line. An equilibrium is efficient only if its cutoff coincides with the boundary of that set.

The three figures (examples checked in `check_E3.py`; blue r, green $E[\theta \mid \theta \le \hat\theta]$, dashed 45°):



- *Fig. 13.B.1* (left): $\theta \sim U[1,2]$, $r(\theta) = 0.9\,\theta \le \theta$. All types should work, but the unique equilibrium is the crossing $\hat\theta = \frac{5}{4}$ (wage $\frac{9}{8}$); here $r(\bar\theta) = 1.8 > E[\theta] = 1.5$, so the all-work corner fails: the best workers are driven out.
- *Fig. 13.B.2* (middle): $\theta \sim U[0,2]$, $r(\theta) = 0.8\,\theta$, $\alpha > \frac{1}{2}$: $E[\theta \mid \theta \le \hat\theta] = \hat\theta/2 < r(\hat\theta)$ for every $\hat\theta > 0$; the only equilibrium is the corner $\hat\theta = \underline\theta = 0$, with a null set employed: complete market failure although everybody should work ($r(\theta) < \theta$).
- *Fig. 13.B.3* (right): a density with two bumps ($\theta \in [0,1]$, mass $\frac{1}{20}, \frac{5}{20}, \frac{1}{20}, \frac{13}{20}$ on $[0, \frac{1}{5}], [\frac{1}{5}, \frac{3}{5}], [\frac{3}{5}, \frac{4}{5}], [\frac{4}{5}, 1]$) and $r(\theta) = 0.6\,\theta$: the curves cross three times, at $\hat\theta \approx 0.262$, 0.458, 0.834, with wages 0.157, 0.275, 0.501. Wages are Pareto ranked (firms earn 0 everywhere, workers gain with w): the high-wage equilibrium dominates, the low ones are coordination failures.

(d) Interpretation. The three curves summarize the model: efficiency is read against the 45° line (types worth more at the firm than at home), the equilibrium against the expected-productivity curve $E[\theta \mid \theta \le \hat\theta]$, which lies below the 45° line because only the worse types accept low wages: adverse selection.

(e) Verification. `check_E3.py`: sympy solves the crossing for Fig. B.1 ($\hat\theta = \frac{5}{4}$, $E = r = \frac{9}{8}$) and shows the $\alpha > \frac{1}{2}$ case of Fig. B.2 has no positive root; it solves the piecewise quadratic equations for Fig. B.3

(three roots, increasing wages), checks that $E[\theta \mid \theta \leq \hat{\theta}]$ is increasing, and writes `fig-mwg13b.tex`. MWG's solutions manual contains the figures; no numbers to compare.

Answer. Equilibria = crossings of $r(\hat{\theta})$ with $E[\theta \mid \theta \leq \hat{\theta}]$ (plus the corners $\hat{\theta} = \bar{\theta}$ if $r(\bar{\theta}) \leq E[\theta]$ and $\hat{\theta} = \underline{\theta}$); the Pareto optimal set of workers is where $r(\hat{\theta}) \leq \hat{\theta}$ (below the 45° line). Fig. B.1: one interior crossing below the efficient set; B.2: only $\hat{\theta} = \underline{\theta}$; B.3: three crossings.

Item 20 (item 20 of the combined list): MWG Exercise 13.B.2 Advanced · difficulty 4/5

Source: MWG Exercise 13.B.2, page 473. The statement is reproduced verbatim in the list (item 20); it is not repeated here.

(a) What is asked. With $r(\cdot)$ continuous and strictly increasing, $r(\theta) > \theta$ for $\theta > \theta^\circ$ and $r(\theta) < \theta$ for $\theta < \theta^\circ$ (the exercise's $\hat{\theta}$, renamed to avoid clashing with the cutoff), and $f > 0$: show that any competitive equilibrium with unobservable types is Pareto inefficient.

(b) Method. The Pareto optimal allocation maximizes aggregate surplus (quasilinear preferences): workers with $r(\theta) \leq \theta$ work, i.e. $\theta \leq \theta^\circ$ (by continuity $r(\theta^\circ) = \theta^\circ$) (MWG pp. 438–439). A competitive equilibrium has a wage w and an accepting set $\{\theta : r(\theta) \leq w\} = [\underline{\theta}, \hat{\theta}]$ with $w = E[\theta \mid \theta \leq \hat{\theta}]$ (Def. 13.B.1). Show that every equilibrium has $\hat{\theta} < \theta^\circ$ and compare the two sets.

(c) Solution. Because $f > 0$, $E[\theta \mid \theta \leq \hat{\theta}] < \hat{\theta}$ for every $\hat{\theta} > \underline{\theta}$. Consider the possible equilibria.

- *Interior cutoff* $\hat{\theta} \in (\underline{\theta}, \bar{\theta})$, $w = r(\hat{\theta})$: the break-even condition gives $r(\hat{\theta}) = E[\theta \mid \theta \leq \hat{\theta}] < \hat{\theta}$, hence $r(\hat{\theta}) < \hat{\theta}$, hence $\hat{\theta} < \theta^\circ$ (since $r \geq \theta$ for $\theta \geq \theta^\circ$).
- *Everybody works,* $\hat{\theta} = \bar{\theta}$, $w = E[\theta]$: needs $r(\bar{\theta}) \leq E[\theta]$; but $\bar{\theta} > \theta^\circ$ gives $r(\bar{\theta}) > \bar{\theta} > E[\theta]$. Impossible.
- *Only the lowest type (a null set) or nobody works:* the employed set has measure zero.

So in every equilibrium the set of workers employed by firms is $[\underline{\theta}, \hat{\theta}]$ with $\hat{\theta} < \theta^\circ$ (or a null set), while efficiency requires $[\underline{\theta}, \theta^\circ]$. The types in $(\hat{\theta}, \theta^\circ)$, a set of positive measure because $f > 0$, are worth more at a firm ($\theta > r(\theta)$) but stay at home. Aggregate surplus falls short of the maximum by

$$N \int_{\hat{\theta}}^{\theta^\circ} \big(\theta - r(\theta)\big) f(\theta)\, d\theta > 0.$$

With quasilinear utility, transfers can redistribute any surplus gain, so some allocation makes everybody weakly better off and somebody strictly: the equilibrium is not Pareto optimal.

Example. $\theta \sim U[1, 2]$, $r(\theta) = 2\theta - \frac{3}{2}$ (so $\theta^\circ = \frac{3}{2}$). The crossing $r(\hat{\theta}) = \frac{1+\hat{\theta}}{2}$ gives $\hat{\theta} = \frac{4}{3} < \frac{3}{2}$, wage $\frac{7}{6}$; the types in $(\frac{4}{3}, \frac{3}{2})$ are lost, and the surplus loss per worker is $\int_{4/3}^{3/2} (\frac{3}{2} - \theta)\, d\theta = \frac{1}{72}$.

(d) Interpretation. The failure is driven by adverse selection: the wage can only equal the *average* productivity of the accepting set, which is below the productivity of the marginal worker, so the marginal worker is never attracted. The situation is the opposite of Fig. 13.B.1–13.B.3 (there all types should work); here only those below θ° should, and still too few do.

(e) Verification. `check_E3.py`: the example (root $\frac{4}{3}$, loss $\frac{1}{72}$, corners fail); and 200 random draws of a positive piecewise-constant density and a linear increasing r crossing θ once: every interior crossing of r and $E[\theta \mid \theta \leq \hat{\theta}]$ lies strictly below θ°, and the all-work corner fails. MWG's manual had no OCR text for 13.B.2.

Answer. Every equilibrium has cutoff $\hat{\theta} < \theta^\circ$ (or employs a null set), so the types in $(\hat{\theta}, \theta^\circ)$ with $\theta > r(\theta)$ do not work at a firm; aggregate surplus is strictly below the maximum, hence the equilibrium is Pareto inefficient.

Item 21 (item 21 of the combined list): Tadelis Exercise 12.7 Advanced · difficulty 4/5

Source: Tadelis Exercise 12.7, page 268. The statement is reproduced verbatim in the list (item 21); it is not repeated here.

(a) What is asked. House swap with private values $v_i \sim U[0,1]$ (house i is worth v_i to its owner and $\frac{3}{2}v_i$ to the other): (a) a threshold BNE; (b) the change if the other's value is $\frac{5}{2}v_i$; (c) why every BNE is a threshold equilibrium.

(b) Method. Player i announces "exchange" iff $v_i \le \theta_i$. An exchange takes place only if both agree, so i's decision matters only when j agrees; compute i's expected gain from agreeing conditional on that event (Tadelis §12.3, pp. 258–261, same adverse-selection logic).

(c) Solution. **(a)** Suppose j agrees iff $v_j \le \theta_j$. Then $\Pr(j \text{ agrees}) = \theta_j$ and, when j agrees, $E[v_j] = \theta_j/2$. If i agrees, he gets $\frac{3}{2}v_j$ when j agrees and keeps v_i otherwise; if he refuses he keeps v_i. The gain from agreeing is

$$\theta_j\Big(\tfrac{3}{2}\cdot\frac{\theta_j}{2} - v_i\Big) \ge 0 \iff v_i \le \tfrac{3}{4}\,\theta_j.$$

This is decreasing in v_i: a threshold rule with $\theta_i = \frac{3}{4}\theta_j$. Simultaneously $\theta_j = \frac{3}{4}\theta_i$, so $\theta_i = \frac{9}{16}\theta_i$ and $\theta_1 = \theta_2 = 0$: the only threshold equilibrium has no one trading (agreeing has probability 0). The reason is adverse selection: a rival who agrees has $E[v_j] = \theta_j/2$, so the house received is worth only $\frac{3}{4}\theta_j < \theta_j$ on average, and each player's cutoff is below the other's.

(b) With $\frac{5}{2}v_j$ the gain is $\theta_j(\frac{5}{2}\cdot\frac{\theta_j}{2} - v_i)$, so $\theta_i = \min\{1, \frac{5}{4}\theta_j\}$. Solutions: $\theta_1 = \theta_2 = 0$, and $\theta_1 = \theta_2 = 1$ (everyone agrees: $\frac{5}{4}\cdot 1 \ge 1$). Interior solutions require $\theta = \frac{25}{16}\theta$, so none with $\theta > 0$. There are two threshold equilibria: no trade, and trade for every pair of types (the swap is then mutually beneficial in expectation: $E[\frac{5}{2}v_j] = \frac{5}{4} > \frac{1}{2} = E[v_i]$).

(c) For a fixed strategy of j with $\Pr(j \text{ agrees}) > 0$, i's gain from agreeing is $\Pr(j \text{ agrees})\big(kE[v_j \mid j \text{ agrees}] - v_i\big)$, a *decreasing linear function* of v_i: i agrees exactly for v_i below a cutoff c. So i's best reply (and hence i's strategy in any BNE) is a threshold rule. The exception is a j who never agrees: then i's choice is irrelevant and any rule is a best reply, but j's own behavior (never agreeing) is the threshold $\theta_j = 0$.

(d) Interpretation. (a) is Akerlof's lemons logic with two-sided private information: gains from trade exist for every pair ($\frac{3}{2}v_j > v_j$) and still no trade survives, because agreeing is informative about the value of the house one receives. (b) Increasing the gain from the swap supports a full-trade equilibrium as well, with no-trade remaining another equilibrium.

(e) Verification. `check_E3.py`: sympy checks $E[v_j \mid v_j \le \theta_j] = \theta_j/2$; scanning a grid of thresholds for the fixed points of $\theta \mapsto \min\{1, k\theta/2\}$ returns $\{(0,0)\}$ for $k = \frac{3}{2}$ and $\{(0,0),(1,1)\}$ for $k = \frac{5}{2}$; checks that $k/2 \cdot k/2 \ne 1$ (no interior solutions); and that the gain function is monotone decreasing in v_i. Tadelis gives no answer.

Answer. (a) The only threshold equilibrium is $\theta_1 = \theta_2 = 0$ (no trade). (b) Thresholds $\theta_i = \min\{1, \frac{5}{4}\theta_j\}$: two equilibria, $(0,0)$ and $(1,1)$ (full trade). (c) The gain from agreeing is decreasing and linear in v_i, so best replies are cutoff rules.

Item 22 (item 22 of the combined list): Gibbons Problem 3.8 Advanced · difficulty 5/5

Source: Gibbons Problem 3.8, page 171–172. The statement is reproduced verbatim in the list (item 22); it is not repeated here.

(a) What is asked. Firm (marginal product m) and worker (outside opportunity v), m, v independent $U[0,1]$: expected payoffs in the linear equilibrium of the double auction; then Game I (fixed wage w, each side accepts or rejects) and Game II (the firm offers a wage): equilibrium, trading region, and the sum of expected payoffs.

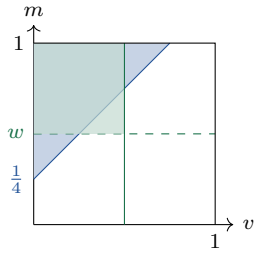
(b) Method. Gibbons §3.2.C (pp. 158–163). Trade gives the firm $m - w$ and the worker w; no trade gives 0 and v. Because v is the worker's payoff without trade, the *sum of expected payoffs* is $E[v] + E[(m-v)\mathbf{1}_{\text{trade}}] = \frac{1}{2} + (\text{gains from trade})$.

(c) Solution. Double auction. Worker (seller) names p_s, firm (buyer) names p_b; trade at $(p_b + p_s)/2$ iff $p_b \geq p_s$. With linear strategies $p_b = a_b + c\,m$, $p_s = a_s + c\,v$ the firm's FOC gives $p_b = \frac{2}{3}m + \frac{1}{3}a_s$ and the worker's $p_s = \frac{2}{3}v + \frac{1}{3}(a_b + c)$; so $c = \frac{2}{3}$, $a_b = \frac{1}{12}$, $a_s = \frac{1}{4}$:

$$p_b(m) = \tfrac{2}{3}m + \tfrac{1}{12}, \qquad p_s(v) = \tfrac{2}{3}v + \tfrac{1}{4}, \qquad \text{trade} \iff p_b \geq p_s \iff m \geq v + \tfrac{1}{4}.$$

Trade probability $\frac{9}{32}$; gains from trade $E[(m-v)\mathbf{1}] = \frac{9}{64}$; price $w = \frac{m+v}{3} + \frac{1}{6}$. Expected payoffs: firm $E[(m-w)\mathbf{1}] = \frac{9}{128}$, worker $\frac{1}{2} + E[(w-v)\mathbf{1}] = \frac{73}{128}$; sum $\frac{82}{128} = \frac{41}{64}$ (gains $\frac{9}{64} \approx 0.141$).

Game I (fixed w). After learning their information the firm accepts iff $m \geq w$ and the worker iff $w \geq v$ (accepting weakly dominates; the trade-never equilibrium in which both always reject is a Nash equilibrium too, but uses weakly dominated actions). Trade occurs iff $m \geq w \geq v$: the rectangle $[0,w] \times [w,1]$ in (v,m) space, versus the double auction's triangle $m \geq v + \frac{1}{4}$.



Trading regions in (v,m) space: blue triangle = double auction; green rectangle = Game I for $w = \frac{1}{2}$ (the two overlap).

Gains from trade at wage w: $P(m \geq w)P(v \leq w)\big[E(m \mid m \geq w) - E(v \mid v \leq w)\big] = (1-w)\,w \cdot \frac{1}{2}$, maximized at $w = \frac{1}{2}$ with value $\frac{1}{8}$, so the maximized sum of expected payoffs is $\frac{1}{2} + \frac{1}{8} = \frac{5}{8}$. At $w = \frac{1}{2}$ the firm earns $\frac{1}{16}$ and the worker $\frac{1}{2} + \frac{1}{16}$. This is below the double auction's $\frac{9}{64}$ in gains (and $\frac{41}{64}$ in sum).

Game II (the firm offers). The worker accepts w iff $w \geq v$, with probability w. The firm with marginal product m maximizes $(m-w)\,w$: $w = m/2$ (the payoff $m^2/4$). Trade occurs iff $v \leq m/2$. Gains from trade $\int_0^1 \int_0^{m/2} (m-v)\,dv\,dm = \int_0^1 \frac{3}{8}m^2 dm = \frac{1}{8}$; firm $\frac{1}{12}$, worker $\frac{1}{2} + \frac{1}{24}$; sum $\frac{5}{8}$.

(d) Interpretation. The double auction is best (gains $\frac{9}{64} > \frac{1}{8}$), though not efficient (first best $E[(m-v)^+] = \frac{1}{6}$): its regions of trade are not rectangular, so it adapts the trade rule to both types. Game II (informed principal) yields the same $\frac{1}{8}$ as the best fixed wage.

(e) Verification. `check_E3.py` (sympy): the linear equilibrium coefficients and best-reply FOC; the trade condition $m \geq v + \frac{1}{4}$; the integrals $\frac{9}{32}$, $\frac{9}{64}$, $\frac{9}{128}$, $\frac{73}{128}$, $\frac{41}{64}$; Game I gains $w(1-w)/2$, firm $w(1-w)^2/2$, worker $\frac{1}{2} + (1-w)w^2/2$, the maximum at $w = \frac{1}{2}$; Game II offer $m/2$, gains $\frac{1}{8}$, payoffs $\frac{1}{12}, \frac{1}{2} + \frac{1}{24}$. Gibbons gives no answer.

Answer. Double auction: firm $\frac{9}{128}$, worker $\frac{73}{128}$, sum $\frac{41}{64}$ (gains $\frac{9}{64}$). Game I: trade iff $m \geq w \geq v$; best $w = \frac{1}{2}$, sum $\frac{5}{8}$ (gains $\frac{1}{8}$). Game II: the firm offers $w = m/2$; trade iff $v \leq m/2$; sum $\frac{5}{8}$ (gains $\frac{1}{8}$).

Item 23 (item 23 of the combined list): Tadelis Exercise 13.5 Advanced · difficulty 5/5

Source: Tadelis Exercise 13.5, page 286. The statement is reproduced verbatim in the list (item 23); it is not repeated here.

(a) What is asked. First-price auction, two bidders, values $\theta_i \in \{4, 8\}$ equiprobable: (a) the low type bids 4; (b) the high type never bids above 6; (c) check the proposed equilibrium; (d) revenue against the English auction.

(b) Method. Symmetric BNE, ties broken by a fair coin. Winning probabilities are weakly increasing in the bid; a mixed strategy needs equal payoffs on its support (Tadelis §6.3, §13.1.3). Let $W(b)$ be the probability that a bid b wins.

(c) Solution. **(a)** A low type never bids above 4 (winning there pays more than 4; bid 4 already guarantees at least 0), and the bid 4 earns exactly 0. *No atom below* 4. Suppose a bid $\beta < 4$ is used with positive probability. A rival low type then also bids β with positive probability, so β ties with positive probability. Raising the bid to $\beta + \varepsilon$ turns each such tie into a sure win, a discrete gain in W, at cost at most ε; so for small ε the deviation pays, contradicting optimality. This rules out a pure $b_L < 4$ and any mixture over two different bids below 4. *No atomless mixing below* 4. If the low type's support were an atomless set below 4, with infimum $\underline{b} < 4$ and supremum $\bar{b} \le 4$, the lowest bids win with probability close to 0 (they can beat only rival low types bidding even lower), so their payoff tends to 0, whereas $\bar{b} < 4$ beats every rival low type and earns at least $\frac{1}{2}(4 - \bar{b}) > 0$; the two cannot be equal. If $\bar{b} = 4$ the same argument applies at $\underline{b}$. Hence the low type's bid is the pure bid $b_L = 4$.

(b) Given $b_L = 4$, a high type bidding $4 + \varepsilon$ beats every rival low type (probability $\frac{1}{2}$), so his payoff is at least $\frac{1}{2}(4 - \varepsilon) \to 2$. A bid $b > 6$ earns at most $8 - b < 2$ (the price is at most b and winning probability is at most 1), which is strictly less than $\frac{1}{2}(4 - \varepsilon)$ for ε small. So no bid above 6 is optimal.

(c) *The uniform distribution on* $[4, 6]$ *is not an equilibrium strategy of the high type; the equilibrium has cdf* $G(b) = \dfrac{b-4}{8-b}$ *on* $[4, 6]$. A high type bidding $b \in (4, 6]$ wins against all low types (probability $\frac{1}{2}$) and against a high type bidding below b (probability $\frac{1}{2}G(b)$):

$$U_H(b) = (8 - b)\Big(\tfrac{1}{2} + \tfrac{1}{2}G(b)\Big).$$

Mixing requires U_H constant on the support. As $b \downarrow 4$, $U_H \to 4 \cdot \frac{1}{2} = 2$ (a bid exactly 4 ties with low types and earns only 1), so $U_H \equiv 2$ gives $1 + G(b) = \dfrac{4}{8-b}$, i.e. $G(b) = \dfrac{b-4}{8-b}$, with $G(4) = 0$, $G(6) = 1$ and density $\dfrac{4}{(8-b)^2}$. With the *uniform* $G(b) = (b-4)/2$ the payoff is $U_H(b) = \frac{(8-b)(b-2)}{4}$, equal to 2 at $b = 4^+$ and $b = 6$ but $\frac{9}{4}$ at $b = 5$: the high type would deviate to a pure bid near 5, so uniform mixing is not a BNE (the book's wording in (c) is imprecise on this point). With the correct G: the high type is indifferent over $[4, 6]$ (payoff 2), strictly worse above 6 ($8 - b < 2$) and strictly worse below 4 (payoff 0); the low type earns 0 at $b_L = 4$ and $(4 - b)W(b) < 0$ at any $b > 4$. So $b_L = 4$ and $b_H \sim G$ form a BNE.

(d) Revenue. Both low ($\frac{1}{4}$): the price is 4. One high ($\frac{1}{2}$): the high type wins and pays $b_H \sim G$, $E[b_H] = \int_4^6 \frac{4b}{(8-b)^2} db$. Both high ($\frac{1}{4}$): the maximum of two draws, cdf G^2. Evaluating,

$$\text{Rev} = \tfrac{1}{4} \cdot 4 + \tfrac{1}{2}E[b_H] + \tfrac{1}{4}E[\max] = 5.$$

The English auction gives the second-highest value: 8 if both are high ($\frac{1}{4}$), else 4: $\frac{1}{4} \cdot 8 + \frac{3}{4} \cdot 4 = 5$. The two are equal: revenue equivalence. (With the uniform on $[4, 6]$ one would get $\frac{29}{6} < 5$, another sign that it is not the equilibrium.)

(d) Interpretation. With only two values, the high type must randomize to avoid being undercut by a pure bid; the support starts at the low type's value because bidding below it never wins against a low type's bid of 4, and the shading is exactly enough to make revenue equal to the English auction's.

(e) Verification. `check_E3.py`: sympy verifies $G(4) = 0$, $G(6) = 1$, $U_H \equiv 2$ on $(4, 6]$; shows $U_H^{\text{unif}}(5) = \frac{9}{4} > 2$; integrates $E[b_H]$ and $E[\max]$ to obtain revenue exactly $5 =$ English revenue; and the low type's strictly negative payoff from bids above 4. Tadelis gives no answer.

Answer. (a) $b_L = 4$. (b) No high bid above 6. (c) The high type mixes on $[4, 6]$ with cdf $G(b) = \frac{b-4}{8-b}$ (the uniform fails: $U_H(5) = \frac{9}{4} > 2$). (d) Revenue 5, equal to the English auction's $\frac{1}{4} \cdot 8 + \frac{3}{4} \cdot 4 = 5$.

Item 24 (item 24 of the combined list): Tadelis Exercise 13.10 Advanced · difficulty 4/5

Source: Tadelis Exercise 13.10, page 287. The statement is reproduced verbatim in the list (item 24); it is not repeated here.

(a) What is asked. Two bidders, first-price auction, value $\theta_1 + \theta_2$ with private signals $\theta_i \sim U[0,1]$: (a) a symmetric BNE $s(\theta_i) = a + b\theta_i$; (b) the expected payoff of type θ_i; (c) compare the bid with item 16.

(b) Method. Common values: winning is informative about the rival's signal (the winner's curse; Tadelis §13.2, pp. 282–284). Let the rival bid $a + b\theta_j$ with $b > 0$; a bid x wins iff $\theta_j < t \equiv (x-a)/b$, and then $E[\theta_j \mid \theta_j < t] = t/2$.

(c) Solution. **(a)** Given signal θ_i and a bid $x = a + bt$,

$$EU_i = \underbrace{t}_{\Pr(\text{win})} \Big(\theta_i + \frac{t}{2} - a - bt\Big).$$

FOC in t: $\theta_i - a + t - 2bt = 0$, so $t = \dfrac{\theta_i - a}{2b-1}$ (second derivative $1 - 2b < 0$ needs $b > \frac{1}{2}$). In a symmetric equilibrium the bid of type θ_i beats exactly the rival types below θ_i, i.e. $t = \theta_i$ for all θ_i:

$$(2b-1)\theta_i = \theta_i - a \;\; \forall \theta_i \;\Rightarrow\; 2b - 1 = 1,\; a = 0 \;\Rightarrow\; b = 1,\; a = 0.$$

So $s^*(\theta_i) = \theta_i$: bid one's own signal. Conditional on winning the item is worth $\theta_i + \theta_i/2$ (the rival's signal is lower), and the bid shades that by $\theta_i/2$.

(b) With $t = \theta_i$: $EU_i^* = \theta_i\big(\theta_i + \frac{\theta_i}{2} - \theta_i\big) = \dfrac{\theta_i^2}{2}$.

(c) Item 16 (private values $0.5 + \theta_i$) has $s(\theta_i) = \frac{1}{2} + \frac{1}{2}\theta_i$. Then

$$\tfrac{1}{2} + \tfrac{1}{2}\theta_i - \theta_i = \tfrac{1}{2}(1 - \theta_i) > 0 \quad \text{for } \theta_i < 1,$$

so the common-value bid is strictly lower for every $\theta_i < 1$ (they coincide at $\theta_i = 1$, a single type). Winner's curse: conditional on winning, the rival's signal is lower than θ_i, so the item is worth less than $0.5 + \theta_i$; bidders shade more.

(d) Interpretation. Both exercises give the same expected payoff $\theta_i^2/2$, but with common values the bid is lower because winning is bad news about the rival's signal.

(e) Verification. `check_E3.py`: sympy solves the FOC, finds $(a, b) = (0, 1)$ by matching, checks strict concavity (-1 at $b = 1$) and the payoff $\theta^2/2$; scipy maximizes EU_i against $s(\theta) = \theta$ at $\theta = 0.1, 0.4, 0.8$ and returns bids θ and value $\theta^2/2$; compares with $\frac{1}{2} + \frac{1}{2}\theta$. Tadelis gives no answer.

Answer. (a) $s^*(\theta_i) = \theta_i$ ($a = 0$, $b = 1$). (b) $EU_i^*(\theta_i) = \theta_i^2/2$. (c) The bid is lower than item 16's $\frac{1}{2} + \frac{1}{2}\theta_i$ by $\frac{1}{2}(1 - \theta_i)$, strictly for $\theta_i < 1$ (winner's curse).